

3.1**ignition source**

source of energy that initiates combustion

[SOURCE: ISO 13943:2017, 3.219]

3.2**char**

carbonaceous residue resulting from pyrolysis or incomplete combustion

[SOURCE: ISO 13943:2017, 3.47]

3.3**flame spread**

propagation of a flame front

[SOURCE: ISO 13943:2017, 3.168]

4 Test apparatus**4.1 General**

The apparatus specified in IEC 60332-3-10 shall be used.

4.2 Ignition source

The ignition source shall be one or two ribbon-type propane gas burners as specified in IEC 60332-3-10.

A single burner shall be used with the standard ladder and two burners with the wide ladder.

5 Test procedure**5.1 Test sample**

The test sample shall comprise a number of test pieces of cable from the same production length, each having a minimum length of 3,5 m.

The total number of test pieces in the test sample shall be that number required to provide a nominal total volume of non-metallic material of 7 l/m of test sample.

The test sample for type approval shall be in accordance with Annex A.

The test pieces forming the test sample shall be conditioned at a temperature of $(20 \pm 10) ^\circ\text{C}$ for at least 16 h before commencing the test. The test pieces shall be dry.

5.2 Determination of the number of test pieces

In order to calculate the appropriate number of test pieces, it is necessary to determine the volume per metre of non-metallic material of one test piece.

A length of cable which shall be not less than 0,3 m long is carefully cut to ensure that the surfaces are at right angles to the cable axis, thus enabling precise measurements of its length.

The density of each non-metallic component (including cellular material) shall be measured in an appropriate way, for example according to IEC 60811-606, in order to obtain values expressed to the second decimal place.

Each non-metallic material C_i shall be removed from the test piece and weighed. Any non-metallic material making up less than 5 % of the total non-metallic mass of the test piece shall be assumed to have a density of 1,0 kg/dm³.

Where semi-conducting screens cannot be removed from the insulating material, the components may be considered as one for the purpose of measuring their mass and density.

The volume V_i (litres per metre of cable) of each non-metallic material C_i is calculated as follows:

$$V_i = \frac{M_i}{\rho_i \times l}$$

where

M_i is the mass of the component C_i (kg);

ρ_i is the density of the component C_i (kg/dm³);

l is the length of the test piece of cable (m).

The total volume, V , of the non-metallic materials contained in 1 m of cable is equal to the sum of the individual volumes V_1 , V_2 , etc.

The number of test pieces to be mounted is obtained by taking the closest integer (0,5 and above corresponding to 1) of the ratio of the volume per metre specified in 5.1 and the total volume, V , of non-metallic material per metre of cable, subject to a minimum number of two test pieces.

5.3 Mounting of the test sample

5.3.1 Cables having at least one conductor above 35 mm²

For cables having at least one conductor with a cross-section exceeding 35 mm², each test piece shall be attached individually to each rung of the ladder by means of metal wire (steel or copper). For cables up to and including 50 mm diameter, use wire between 0,5 mm and 1,0 mm in diameter. For cables above 50 mm diameter, use wire between 1,0 mm and 2,5 mm in diameter.

Test pieces shall be attached to the front of the ladder in a single layer with a space between each test piece of 0,5 times the cable diameter but not exceeding 20 mm. The ladder may be either standard or wide depending on whichever is necessary to ensure that there shall be a minimum distance of 50 mm between the edge of the test sample and the inside of the ladder uprights. Where it is not possible to ensure that there shall be a minimum distance of 50 mm between the edge of the test sample and the inside of the standard ladder, the wide ladder shall be used.

The maximum width of the test sample for the standard ladder shall be 300 mm and for the wide ladder 600 mm (see Figures 1 and 2). When the wide ladder is used, there shall not be more cable than the number necessary to form a single layer of 600 mm width, allowing for a space between each cable of either half the cable diameter or 20 mm, whichever is the smaller.

When mounting the test pieces, the first test piece shall be positioned approximately in the centre of the ladder and further test pieces added on either side so that the whole array of test pieces is approximately centred on the ladder.

Mount all test pieces on the ladder with the bending (due to bending of the cable on the reel), if any, in the same direction, towards the back side of the test chamber as much as possible.

To achieve straight test pieces between the steel rungs below and above the burner position, it is permitted to fix the test pieces to an additional horizontal support 200 mm to 300 mm lower than the steel rung below the burner position.

NOTE The objective is to improve the repeatability and reproducibility of the test.

5.3.2 Cables having conductors 35 mm² and below and optical cables

For cables having all conductors with cross-sections of 35 mm² or smaller and optical cables, each test piece shall be attached, either individually or as part of an array, to each rung of the ladder by means of metal wire (steel or copper) between 0,5 mm and 1,0 mm in diameter.

Test pieces shall be attached to the front of the standard ladder in touching formation in one or more layers up to a maximum total width of 300 mm. There shall be a minimum distance of 50 mm between the edge of the test sample and the inside of the ladder uprights.

When mounting the test pieces, the first test piece or array of test pieces shall be positioned approximately in the centre of the ladder and further test pieces or arrays added on either side so that the test sample is approximately centred on the ladder.

Mount all test pieces on the ladder with the bending (due to bending of the cable on the reel), if any, in the same direction, towards the back side of the test chamber as much as possible.

To achieve straight test pieces between the steel rungs below and above the burner position, it is permitted to fix the test pieces to an additional horizontal support 200 mm to 300 mm lower than the steel rung below the burner position.

In order to ensure that the test pieces are straight on the ladder, the test pieces shall be maintained under tension manually while attaching the test pieces to the rung of the ladder by means of a metal wire.

NOTE The objective is to improve the repeatability and reproducibility of the test.

If a second (or more) layer of test pieces is required after the full width of the ladder has been utilized for the first (or following) layer, then the first test piece or array of test pieces in the second (or following) layer shall be positioned approximately in the centre of the ladder and further test pieces or arrays added on either side so that the second (or following) layer is approximately centred on the ladder.

If a large number of test pieces is required to make up a test sample, the test pieces may be attached to each rung of the ladder in flat arrays of cables of a maximum width of five test pieces using the specified metal wire. For consistency, it is recommended that adjacent arrays of cables are secured together at every rung to ensure that they are in touching formation (see Figure 3).

5.4 Flame application time

The test flame shall be applied for 40 min, after which it shall be extinguished. The air flow rate through the test chamber shall be maintained until cable burning or glowing has ceased, or until a maximum duration of 1 h, after which any remaining cable burning or glowing shall be extinguished.

6 Evaluation of test results

After all cable burning or glowing has ceased or been extinguished, the test sample shall be wiped clean.

All soot is to be ignored if, when wiped off, the original surface is undamaged. Softening or any deformation of the non-metallic material is also to be ignored. The flame spread shall be measured as the extent of the damage. It shall be measured in metres to two decimal places from the bottom edge of the burner to the onset of char. The onset of char is determined as follows:

press against the cable surface with a sharp object, for example a knife blade. Where the surface changes from a resilient to a brittle (crumbling) surface, this indicates the onset of char.

7 Performance requirements

The performance requirements for a particular type or class of wire or cable should preferably be given in the individual cable standard. In the absence of any given requirement, the recommended performance requirements given in Annex B should be used.

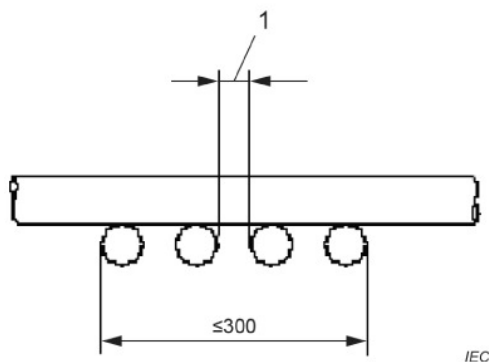
8 Retest procedure

In case of a disputed failure, two further tests shall be undertaken as detailed in Clause 5. The test shall be deemed as satisfactory if both tests meet the stated requirement.

9 Test report

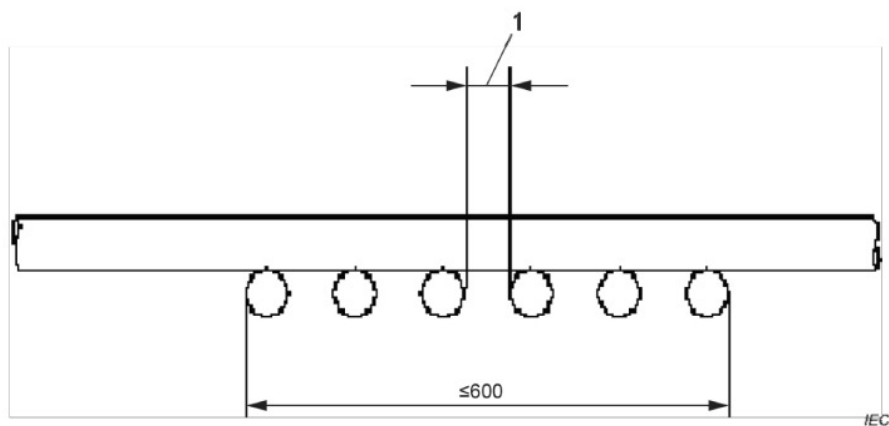
The test report shall include the following information:

- a) full description of the cable tested;
- b) manufacturer of the cable tested;
- c) the part of the standard against which the test was carried out;
- d) the number of test pieces;
- e) the total volume of non-metallic material, per metre of test sample, of the test pieces;
- f) the method of mounting (i.e. spaced or touching);
- g) the number of layers and number of test pieces in each layer;
- h) flame application time (i.e. 40 min);
- i) the number of burners (i.e. one or two);
- j) the extent of damage;
- k) the time to extinction of all burning or glowing.

Dimensions in millimetres**Key**

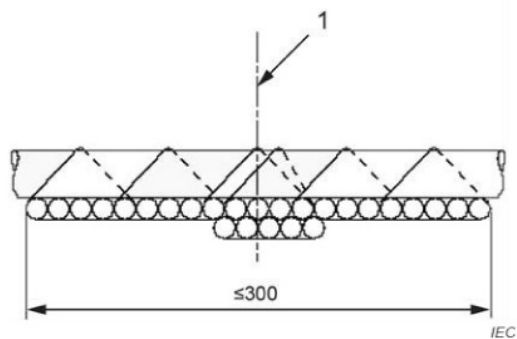
- 1 spacing distance

Figure 1 – Spaced cables mounted on the front side of the standard ladder

Dimensions in millimetres**Key**

- 1 spacing distance

Figure 2 – Spaced cables mounted on the front side of the wide ladder

Dimensions in millimetres**Key**

- 1 centre line of ladder

**Figure 3 – Touching cables mounted on front side of the standard ladder
(arrays of cables in contact)**

Annex A (normative)

Guidance on cable selection for type approval testing

The choice of cable type and conductor cross-section for type approval testing shall be as given in the cable specification, or as agreed between purchaser and manufacturer.

The limited capacity of the ladders requires consideration of the conductor cross-section selected for testing to ensure that the volume of non-metallic material can be accommodated within the specified method of mounting. Therefore, the selection of cables for this document shall comply with the following:

- only the front face of the ladder shall be used;
- for cables having a conductor cross-section greater than 35 mm², there shall not be more cables than the number necessary to form a single layer of 600 mm width, allowing for a space between each cable equal to half the cable diameter but not exceeding 20 mm;
- for cables having a conductor cross-section of 35 mm² or smaller and optical cables, there is no restriction on cable selection;
- the minimum number of test pieces shall be two.

A summary of all conditions for type approval testing to this document is given in Table A.1.

Table A.1 – Summary of test conditions

Category and designation	A	
Range of conductor cross-sections (mm ²)	> 35 ^a	≤ 35 ^b
Non-metallic volume per metre of test sample (l)	7	7
Use of standard ladder, maximum width of test sample: 300 mm		
– number of layers:	1	≥ 1
– number of burners:	1	1
Use of wide ladder, maximum width of test sample: 600 mm		N/A ^c
– number of layers:	1	–
– number of burners:	2	–
Positioning of test pieces	Spaced	Touching
Flame application time (min)	40	40
^a At least one conductor greater than 35 mm ² . ^b No conductor cross-section exceeding 35 mm ² or optical cable. ^c N/A = not applicable.		

Annex B (informative)

Recommended performance requirements

The maximum extent of the charred portion measured on the sample shall not have reached a height exceeding 2,5 m above the bottom edge of the burner.

This recommended performance requirement confirms, on the basis of experience gained, the value previously given in IEC TR 60332-3:1992, 2.8.1.